

30. A fuzzy set A is closed if:

Select one:

- ☐ a. Limit of both sides of membership function are 1
- ☒ b. Limit of both sides of membership function are 0
- ☐ c. Limit of infinity side of membership function is 1, the other limit is 0.
- ☐ d. Limit of infinity side of membership function is 0, the other limit is 1.

44. What Is The Purpose of Aggregation (in Fuzzy Inference System)?

Select one:

- ☐ a. To gather all the different fuzzy set outputs and average them out to get a single value
- ☐ b. To subtract all the output fuzzy set values from the input values
- ☐ c. To gather all the possible inputs and use the average to gain an output
- ☐ d. To gather all the different fuzzy set outputs and combine them into a single fuzzy set outputs

18. Two boxes containing candies are placed on a table. The boxes are labelled B1 and B2. Box B1 contains 7 cinnamon candies and 4 ginger candies. Box B2 contains 3 cinnamon candies and 10 pepper candies.

The boxes are arranged so that the probability of selecting box B1 is $\frac{1}{3}$ and the probability of selecting box B2 is $\frac{2}{3}$. Ahmad is blindfolded and asked to select a candy. He will win a mobile phone if he selects a cinnamon candy.

If he wins a mobile phone, what is the probability that the candy was from the first box?

Select one:

- ☒ a. $\frac{7}{13}$
- ☐ b. $\frac{7}{33}$
- ☐ c. $\frac{6}{33}$
- ☐ d. $\frac{13}{7}$

??

الدكتور جايبر من المنت
والعمل غلط
+ فإل كوين

91
157

13. Which statement is wrong related to a decision tree?

Select one:

- ☐ a. A decision tree is a tree-like structure (a binary tree or a non-binary tree).
- ☒ b. Each non-leaf node represents a test on a feature attribute.
- ☐ c. Each leaf node stores a class.
- ☐ d. Each branch stores the entropy of attribute.

What is the subject of artificial intelligence?

Select one:

- ☐ a. Linguistics
- ☐ b. Mathematics and Physiology
- ☒ c. Comprehensive interdisciplinary and marginal disciplines
- ☐ d. Psychology and Physiology

We have trained a machine learning model to identify that whether the object in an image is a cat. Now we use 200 images to test its performance. Among 200 images, 180 contain cats and 20 do not contain cats.

The identification result of the model is that 140 images contain cats and 60 do not contain cats.

Given that TP (true positive) of the model = 135.

What is the Precision of the model?

Select one:

- ☐ a. 75 %
- ☐ b. 82.5 %
- ☒ c. 96.4 %
- ☐ d. 75.7 %



Transfer learning is:

Select one:

- ☐ a. an area of machine learning concerned with how software agents ought to take actions in an environment in order to maximize the notion of cumulative reward.
- ☐ b. a type of machine learning algorithm used to draw inferences from datasets consisting of input data without labeled responses.
- ☒ c. a machine learning method where a model developed for a task is reused as the starting point for a model on a second task
- ☐ d. the machine learning task of learning a function that maps an input to an output based on example input-output pairs.

44. What Is The Purpose of Aggregation (in Fuzzy Inference System)?

Select one:

- ☒ a. To gather all the different fuzzy set outputs and combine them into a single fuzzy set outputs
- ☐ b. To gather all the possible inputs and use the average to gain an output
- ☐ c. To gather all the different fuzzy set outputs and average them out to get a single value
- ☐ d. To subtract all the output fuzzy set values from the input values

What is the name of this phenomenon that sometimes occurs in machine learning model: "occurs when the training error of the learnt model is small but the test error?" is large

اختراع أحد الخيارات

- a. Overfitting ☒
- b. Underfitting ☐
- c. Generalization ☐
- d. Classification ☐

1. Consider two fuzzy subsets of the set X , $X = \{a, b, c, d, e\}$ referred to as A and B.

$A = \{1/a, 0.3/b, 0.2/c, 0.8/d, 0/e\}$

and $B = \{0.6/a, 0.9/b, 0.1/c, 0.3/d, 0.2/e\}$

Find $A \cap B = ?$

Select one:

- ☒ a. $\{1/a, 0.9/b, 0.2/c, 0.8/d, 2/e\}$
- ☐ b. $\{0/a, 0.7/b, 0.8/c, 0.2/d, 1/e\}$
- ☐ c. $\{0.4/a, 0.1/b, 0.9/c, 0.7/d, 0.8/e\}$
- ☐ d. $\{0.6/a, 0.3/b, 0.1/c, 0.3/d, 0/e\}$

Suppose that Ali can decide to go to work by one of three modes of transportation, car, bus, or train. Because of high traffic, if he decides to go by car, there is a 60 % chance he will be late. If he goes by bus, the probability of being late is 30%. The train is late with a probability of only 15 %.

One day Ali is late. Answer the following questions given that $p(\text{car})=6/12$, $p(\text{bus})=1/12$ $p(\text{train})=5/12$.

Hint: use $h_{\text{MAP}} = \arg \max P(h/D) = \arg \max P(D/h) P(h)$

A. Compute $p(\text{late}/\text{car}) \cdot p(\text{car})$?

B. Compute $p(\text{late}/\text{train}) \cdot p(\text{train})$?

C. Compute $p(\text{late}/\text{bus}) \cdot p(\text{bus})$?

D. What mode of transportation did he use?

We have trained a machine learning model to identify that whether the object in an image is a cat. Now we use 200 images to test its performance. Among 200 images, 180 contain cats and 20 do not contain cats.

The identification result of the model is that 140 images contain cats and 60 do not contain cats.

Given that TP (true positive) of the model = 135.

What is the Precision of the model?

Select one:

- ☐ a. 75 %
- ☒ b. 96.4 %
- ☐ c. 75.7 %
- ☐ d. 82.5 %



31. The support of Fuzzy Set A is the set of all points x in X (is the universe of discourse) such that

Select one:

- ☐ a. Value of membership function of $x = 0$
- ☐ b. Value of membership function of $x = 0.5$
- ☒ c. Value of membership function of $x > 0$
- ☐ d. Value of membership function of $x = 1$

13. Which statement is wrong related to a decision tree?

Select one:

- ☐ a. Each branch stores the entropy of attribute.
- ☒ b. Each non-leaf node represents a test on a feature attribute.
- ☐ c. A decision tree is a tree-like structure (a binary tree or a non-binary tree).
- ☐ d. Each leaf node stores a class.

Consider the training examples shown in the Table below for a binary classification problem

Where a_1 , a_2 and a_3 are the attributes

Instance	a_1	a_2	a_3	Target Class
1	T	T	1.0	+
2	T	T	6.0	+
3	T	F	5.0	-
4	F	F	4.0	+
5	F	T	7.0	-
6	F	T	3.0	-
7	F	F	8.0	-
8	T	F	7.0	+
9	F	T	5.0	-

A. What is the entropy of this collection of training examples

Choose... ▾

0.991

B. What is the information gains of a_1 relative to these training examples?

Choose... ▾

0.229

What is the name of this task that can be solved with machine learning: "In this type of task, the computer program is asked to predict a numerical value given some input."

Select one:

- ☐ a. Crossvalidation
- ☐ b. Classification
- ☒ c. Regression
- ☐ d. Generalization



17. In Naive Bayes classification:

Select one:

- ☐ a. Assumption that all input attributes are related to each other
- ☐ b. Assumption that all input attributes are mutually dependent
- ☐ c. Assumption that all input attributes are conditionally dependent
- ☒ d. Assumption that all input attributes are conditionally independent

37. Defuzzification is done to obtain

Select one:

- ☒ a. Crisp output
- ☐ b. The best rule to follow
- ☐ c. Precise fuzzy value
- ☐ d. None of the above

In unsupervised learning, training set of data includes

Select one:

- ☐ a. Output
- ☒ b. Both input and output
- ☐ c. None
- ☐ d. Input

In supervised learning, training set of data includes

Select one:

- ☐ a. Output
- ☒ b. Both input and output
- ☐ c. None
- ☐ d. input

Two fuzzy sets A and B with membership functions $G(x)$ and $H(x)$, respectively defined as below.

A = Hot Climate with $G(x)$ as the MF.

B = Cold Climate with $H(x)$ as the M.F.

Pleasant (best) climate is given by:

Select one:

- ☐ a. $1 - H(X)$
- ☐ b. $1 - G(x)$
- ☐ c. $\max (G(x), H(x))$
- ☒ d. $\min (G(x), H(x))$



1. Consider two fuzzy subsets of the set X , $X = \{a, b, c, d, e\}$ referred to as A and B .

$$A = \{1/a, 0.3/b, 0.2/c, 0.8/d, 0/e\}$$

$$\text{and } B = \{0.6/a, 0.9/b, 0.1/c, 0.3/d, 0.2/e\}$$

Find $A \cup B = ?$

Select one:

- ☒ a. $\{1/a, 0.9/b, 0.2/c, 0.8/d, 0.2/e\}$
- ☐ b. $\{0/a, 0.7/b, 0.8/c, 0.2/d, 1/e\}$
- ☐ c. $\{0.4/a, 0.1/b, 0.9/c, 0.7/d, 0.8/e\}$
- ☐ d. $\{0.6/a, 0.3/b, 0.1/c, 0.3/d, 0/e\}$

40. What is a membership function of a fuzzy set?

Select one:

- ☒ a. It describes the degree of membership that an object belongs to the set.
- ☐ b. A collection of fuzzy sets
- ☐ c. It is a set of elements that have some common property (such as 'Hot')
- ☐ d. none

35. If x is A then y is B else y is C . The output of the given fuzzy rule is

Select one:

- ☐ a. a fuzzy set
- ☐ b. a crisp set
- ☐ c. a membership function
- ☒ d. a fuzzy relation

32. Which of the following cannot be stated using fuzzy logic?

Select one:

- ☐ a. Height of a person
- ☒ b. Date of birth of a student
- ☐ c. Color of an apple
- ☐ d. Speed of a car

Transfer learning is:

Select one:

- ☐ a. an area of machine learning concerned with how software agents ought to take actions in an environment in order to cumulative reward.
- ☐ b. the machine learning task of learning a function that maps an input to an output based on example input-output pairs.
- ☐ c. a type of machine learning algorithm used to draw inferences from datasets consisting of input data without labeled.
- ☐ d. a machine learning method where a model developed for a task is reused as the starting point for a model on a second task.

ID	Fever	Cough	Breathing issues	Infected
1	NO	NO	NO	NO
2	YES	YES	YES	YES
3	YES	YES	NO	NO
4	YES	NO	YES	YES
5	YES	YES	YES	YES
6	NO	YES	NO	NO
7	YES	NO	YES	YES
8	YES	NO	YES	YES
9	NO	YES	YES	YES
10	YES	YES	NO	YES
11	NO	YES	NO	NO
12	NO	YES	YES	NO
13	NO	YES	YES	NO
14	YES	YES	NO	NO

Using K-NN algorithm compute the probability of a person having a fever (infection rate of fever)

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Hint: Every row in the dataset should contain exactly 5 data after vectorization.

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The boxes are arranged so that the probability of selecting box B1 is $\frac{1}{3}$ and the probability of selecting box B2 is $\frac{2}{3}$. Ahmad is blindfolded and asked to select a candy. He will win a mobile phone if he selects a cinnamon candy.

If he wins a mobile phone, what is the probability that the candy was from the first box?

Select one

- ☐ a. $\frac{13}{7}$
- ☐ b. $\frac{6}{33}$
- ☐ c. $\frac{7}{33}$
- ☒ d. $\frac{7}{13}$



32. Which of the following cannot be stated using fuzzy logic?

Select one:

- ☐ a. Speed of a car
- ☐ b. Height of a person
- ☐ c. Color of an apple
- ☒ d. Date of birth of a student

Consider two fuzzy subsets of the set X , $X = \{a, b, c, d, e\}$ referred to

$A = \{1/a, 0.3/b, 0.2/c, 0.8/d, 0/e\}$

and $B = \{0.6/a, 0.9/b, 0.1/c, 0.3/d, 0.2/e\}$

Find $A' = ?$

Select one:

- ☒ a. $\{0/a, 0.7/b, 0.8/c, 0.2/d, 1/e\}$
- ☐ b. $\{0.6/a, 0.3/b, 0.1/c, 0.3/d, 0/e\}$
- ☐ c. $\{1/a, 0.9/b, 0.2/c, 0.8/d, 2/e\}$
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Two fuzzy sets A and B with membership functions $G(x)$ and $H(x)$, respectively de

A = Hot Climate with $G(x)$ as the MF.

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Pleasant (best) climate is given by:

Select one:

- ☒ a. $\min (G(x), H(x))$
- ☐ b. $1- H(X)$
- ☐ c. $\max (G(x), H(x))$
- ☐ d. $1- G(x)$

16. Previous probabilities in Bayes Theorem that are changed with help of new available information are classified as:

Select one:

- ☒ a. posterior probabilities
- ☐ b. interior probabilities
- ☐ c. dependent probabilities
- ☐ d. independent probabilities



1. Consider two fuzzy subsets of the set X , $X = \{a, b, c, d, e\}$ referred to as A and B .

$$A = \{1/a, 0.3/b, 0.2/c, 0.8/d, 0/e\}$$

$$\text{and } B = \{0.6/a, 0.9/b, 0.1/c, 0.3/d, 0.2/e\}$$

Find $A \cup B = ?$

Select one:

- ☒ a. $\{1/a, 0.9/b, 0.2/c, 0.8/d, 0.2/e\}$
- ☐ b. $\{0.4/a, 0.1/b, 0.9/c, 0.7/d, 0.8/e\}$
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In supervised learning, training set of data includes

Select one:

- ☐ a. Output
- ☒ b. Both input and output
- ☐ c. Input
- ☐ d. None

In machine learning models, the data validation set is used to tune

Select one:

- ☒ a. The model hyper parameters
- ☐ b. The model internal parameters
- ☐ c. underfitting
- ☐ d. overfitting

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- ☐ b. dependent probabilities
- ☐ c. independent probabilities
- ☐ d. posterior probabilities



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Select one:

- ☒ a. a fuzzy set
- ☐ b. a fuzzy relation
- ☐ c. a crisp set
- ☐ d. a membership function

40. What is a membership function of a fuzzy set?

Select one:

- ☒ a. It describes the degree of membership that an object belongs to the set.
- ☐ b. none
- ☐ c. It is a set of elements that have some common property (such as 'Hot')
- ☐ d. A collection of fuzzy sets

1. Consider two fuzzy subsets of the set $X, X = \{a, b, c, d, e\}$ referred to as A and B.

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and $B = \{0.6/a, 0.9/b, 0.1/c, 0.3/d, 0.2/e\}$

Find $A \cup B = ?$

Select one:

- ☐ a. $\{0.5/a, 0.3/b, 0.1/c, 0.3/d, 0/e\}$
- ☐ b. $\{0.4/a, 0.1/b, 0.9/c, 0.2/d, 0.8/e\}$
- ☐ c. $\{0/a, 0.7/b, 0.8/c, 0.2/d, 1/e\}$
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Select one:

- ☒ a. $\min (G(x), H(x))$
- ☐ b. $1 - H(X)$
- ☐ c. $\max (G(x), H(x))$
- ☐ d. $1 - G(x)$

What is the subject of artificial intelligence?

Select one

- ☒ a. Psychology and Physiology
- ☐ b. Mathematics and Physiology
- ☐ c. Linguistics
- ☐ d. Comprehensive interdisciplinary and marginal disciplines

1. Consider two fuzzy subsets of the set X , $X = \{a, b, c, d, e\}$ referred to as A and B

$A = \{1/a, 0.3/b, 0.2/c, 0.8/d, 0/e\}$

and $B = \{0.6/a, 0.9/b, 0.1/c, 0.3/d, 0.2/e\}$

Find $A \cap B = ?$

Select one:

- ☐ a. $\{0.6/a, 0.3/b, 0.1/c, 0.3/d, 0/e\}$
- ☒ b. $\{1/a, 0.9/b, 0.2/c, 0.8/d, 2/e\}$
- ☐ c. $\{0.4/a, 0.1/b, 0.9/c, 0.7/d, 0.8/e\}$
- ☐ d. $\{0/a, 0.7/b, 0.8/c, 0.2/d, 1/e\}$

What is the name of this task that can be solved with machine learning: "In this type of task, the computer program is asked to predict a number given some input."

Select one:

- ☐ a. Generalization
- ☒ b. Regression
- ☐ c. Crossvalidation
- ☐ d. Classification



36. What are the applications of Fuzzy Inference Systems?

Select one:

- ☐ a. Restrict power usage, telephone lines and sort data
- ☒ b. Automatic control, decision analysis and data classification
- ☐ c. Wireless services, fiscal control and printers
- ☐ d. Simulink, boiler and CD recording

Consider two fuzzy subsets of the set $X = \{a, b, c, d, e\}$ referred to as A and B.

$A = (1/a, 0.5/b, 0.2/c, 0.8/d, 0/e)$

and $B = (0.6/a, 0.9/b, 0.1/c, 0.3/d, 0.2/e)$

Find $A \cap B$?

Select one:

- ☐ a. $(0.6/a, 0.9/b, 0.1/c, 0.3/d, 0/e)$
- ☐ b. $(0/a, 0.7/b, 0.8/c, 0.2/d, 1/e)$
- ☐ c. $(1/a, 0.9/b, 0.2/c, 0.8/d, 2/e)$
- ☐ d. $(0.4/a, 0.1/b, 0.9/c, 0.7/d, 0.6/e)$

Regarding entropy which statement is False.

(Hint: S is a collection of positive and negative examples)

Select one:

- ☐ a. Entropy is 1 when S contains an equal number of positive and negative examples
- ☐ b. Entropy is measured in bits.
- ☐ c. Entropy measures the uncertainty.
- ☒ d. Entropy is 0 when S contains an equal number of positive and negative examples

37. Defuzzification is done to obtain

Select one:

- ☒ a. Crisp output
- ☐ b. The best rule to follow
- ☐ c. None of the above
- ☐ d. Precise fuzzy value

We have trained a machine learning model to identify that whether the object in an image is a cat. Now we use 200 images to test its performance. Among 200 images, 180 contain cats and 20 do not contain cats. The identification result of the model is that 140 images contain cats and 60 do not contain cats. Given that TP (true positive) of the model = 135.

What is the Accuracy of the model?

Select one:

- ☐ a. 75.7 %
- ☐ b. 82.5 %
- ☒ c. 96.4 %
- ☐ d. 75 %



14. ID3 (one of the decision tree algorithms) makes use of information gain to arrange the attributes in a tree

Select one

☒ a True

☐ b False

Spaer



Two fuzzy sets A and B with membership functions $G(x)$ and $H(x)$, respectively defined as below.

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- ☒ a. $\min(G(x), H(x))$
- ☐ b. $1 - G(x)$
- ☐ c. $1 - H(x)$
- ☐ d. $\max(G(x), H(x))$

17. In Naive Bayes classification:

Select one:

- ☐ a. Assumption that all input attributes are conditionally dependent
- ☐ b. Assumption that all input attributes are related to each other
- ☐ c. Assumption that all input attributes are mutually dependent
- ☐ d. Assumption that all input attributes are conditionally independent

Consider two fuzzy subsets of the set X , $X = \{a, b, c, d, e\}$ referred to as A and B .

$A = \{1/a, 0.3/b, 0.2/c, 0.8/d, 0/e\}$

and $B = \{0.6/a, 0.9/b, 0.1/c, 0.3/d, 0.2/e\}$

Find $A' = ?$

Select one

- ☒ a. $\{0/a, 0.7/b, 0.8/c, 0.2/d, 1/e\}$
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Suppose that Ali can decide to go to work by one of three modes of transportation: car, bus, or train. Because of high traffic, if he decides to go by car, there is a 60 % chance he will be late. If he goes by bus, the probability of being late is 30%. The train is late with a probability of only 15 %.

One day Ali is late. Answer the following questions given that $p(\text{car}) = 6/12$, $p(\text{bus}) = 1/12$, $p(\text{train}) = 5/12$.

Hint: Use $P_{\text{late}} = \arg \max P(h|D) = \arg \max P(D|h) P(h)$

A. Compute $p(\text{late}|\text{car})/p(\text{car})?$

B. Compute $p(\text{late}|\text{train})/p(\text{train})?$

C. Compute $p(\text{late}|\text{bus})/p(\text{bus})?$

D. What mode of transportation did he use?

1. Consider two fuzzy subsets of the set X , $X = \{a, b, c, d, e\}$ referred to as A and B .

$A = \{1/a, 0.3/b, 0.2/c, 0.8/d, 0/e\}$

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Find $A \cup B = ?$

Select one:

- ☐ a. $\{0/a, 0.7/b, 0.8/c, 0.2/d, 1/e\}$
- ☒ b. $\{1/a, 0.9/b, 0.2/c, 0.8/d, 0.2/e\}$
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- ☐ d. $\{0.6/a, 0.3/b, 0.1/c, 0.3/d, 0/e\}$

15. Decision tree is an algorithm for which machine learning task?

Select one:

- ☐ a. regression
- ☒ b. classification
- ☐ c. clustering
- ☐ d. dimensionality reduction

Consider two fuzzy subsets of the set X , $X = \{a, b, c, d, e\}$ referred to as A and B .

$A = \{1/a, 0.3/b, 0.2/c, 0.8/d, 0/e\}$

and $B = \{0.6/a, 0.9/b, 0.1/c, 0.3/d, 0.2/e\}$

Find $A' = ?$

Select one:

- ☒ a. $c. \{0/a, 0.7/b, 0.8/c, 0.2/d, 1/e\}$
- ☐ b. $b. \{1/a, 0.9/b, 0.2/c, 0.8/d, 2/e\}$
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- ☐ c. Assumption that all input attributes are conditionally dependent
- ☐ d. Assumption that all input attributes are mutually dependent

In unsupervised learning, training set of data includes

Select one:

- ☐ a. Both input and output
- ☒ b. Input
- ☐ c. Output
- ☐ d. None

13. Which statement is wrong related to a decision tree?

Select one:

- ☐ a. Each branch stores the entropy of attribute.
- ☐ b. Each leaf node stores a class.
- ☐ c. A decision tree is a tree-like structure (a binary tree or a non-binary tree).
- ☒ d. Each non-leaf node represents a test on a feature attribute.

31. The support of Fuzzy Set A is the set of all points x in X (is the universe of discourse) such that

Select one:

- ☐ a. Value of membership function of $x = 0$
- ☐ b. Value of membership function of $x = 0.5$
- ☐ c. Value of membership function of $x = 1$
- ☒ d. Value of membership function of $x > 0$

18. Two boxes containing candies are placed on a table. The boxes are labelled B1 and B2. Box B1 contains 7 cinnamon candies and 6 pepper candies. Box B2 contains 3 cinnamon candies and 10 pepper candies.

The boxes are arranged so that the probability of selecting box B1 is $\frac{1}{3}$ and the probability of selecting box B2 is $\frac{2}{3}$. Ahmed is to select a candy. He will win a mobile phone if he selects a cinnamon candy.

If he wins a mobile phone, what is the probability that the candy was from the first box?

Select one:

- ☐ a. $\frac{7}{13}$
- ☒ b. $\frac{7}{33}$
- ☐ c. $\frac{6}{33}$
- ☐ d. $\frac{13}{7}$

SparX
POLYTECHNICS

38. Fuzzification is done to obtain

Select one:

- ☐ a. Crisp output
- ☐ b. The best rule to follow
- ☐ c. None of the above
- ☒ d. Precise fuzzy value

We have trained a machine learning model to identify that whether the object in an image is a cat. Now we use 200 images to test its performance. Among 200 images, 180 contain cats and 20 do not contain cats. The identification result of the model is that 140 images contain cats and 60 do not contain cats. Given that TP (true positive) of the model = 135.

What is the Precision of the model?

Select one:

- ☐ a. 75.7 %
- ☐ b. 75 %
- ☒ c. 96.4 %
- ☐ d. 82.5 %

